UTM CSC343: Introduction to Databases
Midterm Examination October 25, 2002 12:10-1:10

Student Number:________________

Last Name:________________ First Name:________________

This exam consists of 3 double sided pages. Unless otherwise specified, assume that you are using a SQL standard system (ie that has assertions). You are allowed one $8\frac{1}{2} \times 11$ double sided aid sheet.

1. (5 Marks: Medium) Consider the `Beers(bar, beer, price)` relation and the following expression from lecture:

$$\sigma_{beer1 \neq beer}((\rho_{S(bar,beer1,price)}(Beers)) \bowtie Beers)$$

   (a) What is the schema for the this expression (ie. what fields are returned).

   (b) Write an equivalent expression using only $\rho, \pi, \sigma, \times$. Full marks only for the same schema and resulting rows.

For the next questions we will be considering the following VideoStore schema

`Customer(name, address,numRented) Video(title, year, type, length)`
`Rental(name, title, year, rentDate, returned)`

The key for the Customer relation is *name*, for the Video relation is *title, year*, for the Rental table is *name, title, year*. rentDate is the date the video was rented, returned indicates whether the video was returned, length is the length in minutes of the video.

2. (10 Marks: Mixed) Write SQL to create the above relational schema. Include all of the following constraints. Use any reasonable assumption about text field sizes. **Note:**Some of the constraints may not be expressable in postgresql.

   (a) Primary keys as described above.

   (b) Obvious foreign key constraints. Videos can not be deleted if a customer has rented it. Rentals should reflect any changes to Customers or Videos. If a customer is deleted, so is all their rental information.

   (c) Video type is one of *Horror, SciFi, Drama* and must be specified.

   (d) Unless otherwise specified, returned is false.

   (e) numRented is at least 0 and always reflects the number of returned rental items.

(2 continued)

3. (10 Marks: Hard) Consider the following query

```
SELECT C.name, C.address
FROM Customer C
WHERE 3>(SELECT COUNT(DISTINCT V.type)
         FROM Video V NATURAL JOIN Rental R
         WHERE C.name=R.name);
```

   (a) Describe simply (and in english) what this query returns. No marks for things like: *The rows in the customer table for which 3 is greater than the count of the...*

   (b) Write an equivalent SQL query which does not use a sub-select. Your query must return the same rows and have the same schema.

   (c) Write a Relational Algebra expression equivalent to either of the above. For this, you can assume that you can `COUNT(DISTINCT V.type)` etc. in the Relational Algebra.

4. (5 Marks: Easy) Find all customers who have not returned a 3 hour movie. Your query should produce each customers name and address at most one time.

5. (5 Marks: Medium) Create a view named VRC which has, for each video, the number of times it was rented (whether or not it was returned).

6. (5 Marks: Hard) Find the Videos (title, year) which have been rented the most (whether or not it was returned).